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Strategic Prompt Engineering for Enhancing AI-Generated Content in English Language Teaching Empowering EFL Contexts

Aisha Bhatti
https://orcid.org/0000-0001-5037-5597
Prince Sattam Bin Abdulaziz University, Saudi Arabia
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ABSTRACT

Strategic prompt engineering has emerged as a crucial factor in maximizing the academic potential of AI-generated content in empowering English as a foreign language (EFL) context. This study explored the effectiveness of artificial intelligence (AI) tools guided by prompt design in enhancing language teaching. Data for the quantitative analysis was gathered from 77 EFL teachers across institutions in the Kingdom of Saudi Arabia using a structured questionnaire. The responses were analyzed using IBM's Statistical Package for Social Sciences software. Additionally, semi-structured interviews were conducted with 10 participants using NVivo 15 to obtain qualitative insights. Findings indicated that AI-generated content is viewed as a valuable tool; its efficacy is largely dependent on the teacher's ability to craft and refine prompts that align with specific learning aims. Participants reported an increased engagement, particularly in lesson customization and content variety, and emphasized the need for targeted training in prompt engineering. The research highlighted the need for professional development in prompt design and presented practical implications for integrating AI prompt engineering into EFL instruction.

KEYWORDS

AI-Generated Content, Prompt Engineering, English Language, EFL Classrooms, Teacher Perceptions

INTRODUCTION

The integration of artificial intelligence (AI) in education has significantly impacted instructional design in English language teaching (ELT), particularly through the use of tools that support content generation, formative feedback, and personalized learning (Zawacki-Richter et al., 2019). AI-powered tools, such as ChatGPT, Google Bard, and Grammarly, are increasingly being used to assist language instructors in developing lesson plans, generating exercises, and providing instant feedback to students. These tools offer immense potential to enhance both teaching effectiveness and learning outcomes, particularly for learners of English as a foreign language (EFL).

A central aspect of leveraging AI in language education is prompt engineering, which involves crafting precise and contextually appropriate inputs to optimize AI-generated responses. A key component in maximizing the effectiveness of these tools is prompt engineering, the practice of

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designing precise, contextually relevant inputs to elicit high-quality responses from AI models (Bansal, 2024; Ortolan, 2023; Reynolds & McDonell, 2021). However, the effectiveness of AI-generated content largely depends on the quality of the prompts provided by users, highlighting the need for strategic prompt engineering in EFL classrooms. This process, known as prompt-based learning, enables large language models to function more effectively within instructional settings by embedding the task directly into the input. This skill is essential because the quality of AI-generated content is directly influenced by how clearly and strategically the user formulates their query. In pedagogical settings, prompt-based learning refers to embedding instructional tasks directly into prompts, enabling language models to produce more targeted and educationally useful outputs (Gero et al., 2022). For instance, a vague prompt, such as "Explain the past tense," may yield a generic explanation, whereas a well-crafted prompt, like "Provide five examples of past tense verbs in different sentence structures with explanations," would result in a more comprehensive and sound output for teaching purposes.

AI applications in ELT have introduced a range of benefits, including the ability to customize learning experiences to meet diverse student needs (Dugošija, 2024). AI-driven platforms offer a range of instructional benefits. These platforms also enable adaptive learning, where tailored content can improve student proficiency levels, learning styles, and pacing (Aggarwal, 2023; Dutta et al., 2024). Additionally, AI tools support writing enhancement by suggesting grammatical corrections, improving sentence structures, and providing stylistic recommendations, which are particularly useful for non-native English speakers (Wang & Kabilan, 2024). Chatbots offer conversational practice in a low-pressure environment, while well-crafted prompts enhance the relevance, clarity, and coherence of generated content (Belda-Medina & Calvo-Ferrer, 2022; Rosario, 2022; Xiao et al., 2024). Moreover, a recent critical book review by Parviz (2025) on integrating generative AI into English writing classrooms highlighted how AI tools can serve as influential collaborators in enhancing writing processes through adaptable, classroom-ready techniques. The book offers practical frameworks for educators to effectively utilize AI and acknowledge its limitations in addressing significant ethical concerns and issues of fairness. Additionally, this review examined the varied requirements of multilingual learners. The work contributes as an AI facilitator to current disputes on how to support both teachers and learners.

Despite its potential benefits, current challenges in prompt engineering hinder its widespread and practical application in English language classrooms. Many educators lack formal training in AI technologies, including prompt optimization, which can lead to difficulties in generating high-quality teaching content (De la Vall & Araya, 2023; Yang, 2024). Additional concerns include the contextual accuracy of AI-generated responses, alignment with curricular standards, and the depth of pedagogical content (Caines et al., 2023; Warschauer et al., 2023). These challenges highlight an urgent need for teacher competence in strategic prompt design to ensure the responsible and effective integration of AI into ELT.

Although previous studies have explored the general use of AI in education, they have not sufficiently addressed how English language teachers apply prompt engineering in classroom practice. This study aimed to fill that gap by examining how English language teachers perceive and utilize AI-generated content, with a focus on the role of strategic prompt engineering in enhancing the quality of instructional materials. Through a mixed-methods approach involving interviews and questionnaires, the research aimed to identify effective strategies, common challenges, and best practices in prompt design. Ultimately, this study contributes to the evolving field of AI in education by providing a framework for best practices in prompt engineering tailored to ELT.

LITERATURE REVIEW

Recent studies highlight the critical role of prompt engineering, as the strategic formulation of inputs to guide AI-generated responses effectively has emerged as a key factor in enhancing the quality of AI-assisted ELT, particularly in terms of content relevance, learner engagement, and instructional

effectiveness (Mzwri & Turcsányi-Szabo, 2025; Zhao, 2025). In educational settings, AI functions across three primary domains:

- learner-facing AI, designed to support students in their learning process;
- teacher-facing AI, utilized by educators to enhance teaching activities such as grading and lesson planning; and
- system-facing AI, employed by administrative staff to manage and analyze student data efficiently (Pokrivčáková, 2019).

Empirical studies have demonstrated that AI technologies can improve specific language skills, including reading comprehension, repeated language practice, and pronunciation accuracy (Kim, 2019; Noviyanti, 2020; Xu et al., 2019).

The Role of AI in English Language Classrooms

AI is a dynamic and innovative field in ELT (Zhu, 2017). Marr (2018) highlighted AI's potential to foster inclusive classrooms by supporting multilingual environments and accommodating learners with visual or auditory impairments. Building on this inclusive vision, Eslit (2024) emphasized the role of AI in creating accessible and equitable learning experiences through speech-to-text, text-to-speech, and real-time translation technologies. These features not only broaden access but also enhance learner autonomy and engagement.

Several studies have highlighted the benefits of AI in ELT when appropriately applied. Harahap & Fithriani (2024) further reinforced this view by identifying AI as one of the most practical resources for ELT, primarily through personalized and adaptive learning platforms. In the context of language teaching, prompt engineering involves an iterative process in which teachers carefully design, test, and refine the prompts used with AI tools to maximize learning outcomes. Teachers typically begin by formulating an initial prompt aligned with a specific pedagogical goal (e.g., vocabulary expansion, grammar correction, or critical thinking practice). Based on the AI output and student responses, the prompt is then revised—either simplified for clarity, made more specific to guide the AI toward the desired outcome, or expanded to include contextual details that encourage deeper engagement. This cycle of development, revision, and refinement enables teachers to progressively tailor prompts to meet students' needs. AI is among the most practical tools for enhancing English language instruction, particularly through personalized and adaptive learning technologies. These adaptive systems, powered by big data and AI, are now widely available, enabling content to be tailored to individual student needs and thereby increasing engagement and effectiveness. Gawate (2019) emphasized that AI serves as a crucial support system for both students and teachers of the English language, seamlessly integrating language literacy and digital literacy and enhancing global competence in English learning and teaching. Additionally, personalized content plays a vital role in digital learning technology. Big data and AI-powered adaptive systems are already widely accessible. As a globally spoken language with a structured grammatical framework, English poses significant learning challenges for English as a second language/EFL students (Mehrotra, 2019).

Despite these advantages, integrating AI into English language classrooms presents several challenges. On the technical front, limitations in AI-generated content, such as factual inaccuracies, lack of contextual awareness, or cultural misalignment, can hinder its pedagogical effectiveness (Elsayed, 2024; Wagner, 2024). While adaptive systems are widely accessible, their effectiveness depends heavily on the quality of the input provided, highlighting the importance of prompt engineering skills that many teachers have not yet formally developed (Chen et al., 2025). Pedagogically, educators express uncertainty about integrating AI without diminishing communicative competence or critical thinking. Moreover, overreliance on AI tools could risk reducing teacher agency or inadvertently promoting surface-level learning. These issues underscore the need for ongoing teacher training,

critical evaluation of AI outputs, and thoughtful instructional design to ensure AI's role remains supportive rather than substitutive in language education.

Prompt Engineering in Educational Contexts

With the rapid advancement of artificial intelligence (AI) in education, prompt engineering has emerged as a crucial skill for optimizing AI-driven learning experiences. As AI models, particularly large language models, become integral in language education, crafting effective prompts ensures that students receive accurate, context-aware, and engaging responses (Brown et al., 2020). Prompt engineering involves designing precise and structured inputs to guide AI outputs, enhancing their relevance and effectiveness in educational contexts. In English language learning, well-designed prompts can foster reading comprehension, critical thinking, and creative writing by facilitating interactive and engaging AI-driven responses (Tseng & Lin, 2024).

Scholars have proposed varied yet complementary strategies for effective prompt engineering. While Shin et al. (2023) emphasized the importance of clarity and specificity, they provided detailed instructions on the format, length, and complexity of AI-generated responses, thereby significantly enhancing content quality. Wang & Kabilan (2024) emphasized contextual adaptation based on learning objectives and student proficiency levels. Ferrara (2023) introduced iterative refinement, encouraging teachers to revise prompts through trial and error to enhance pedagogical alignment. Complementing these approaches, Muñoz-Basols et al. (2023) promoted the use of guided response structuring, using explicit directives, such as "Explain using examples" or "Compare and contrast concepts," to enhance the coherence and utility of AI responses.

These studies collectively demonstrated that prompt engineering requires a balanced integration of linguistic precision, contextual relevance, and iterative improvement. However, empirical research examining how teachers and learners actually apply these strategies in ELT remains limited. Most existing work is conceptual, highlighting a significant gap in classroom-based evidence. This gap underscores the relevance of the present study, which examined how learners utilize prompt engineering to enhance ELT-related tasks and identified the challenges and opportunities that arise.

Teacher Perceptions of AI Integration in the Classroom

Teacher perceptions play a pivotal role in determining the success of AI integration in ELT, particularly in relation to the use of prompt engineering. A research study conducted by Kumar et al. (2024) suggested that teacher training in prompt engineering significantly enhances the effectiveness of AI-assisted instruction. Teachers familiar with how AI generates responses are better able to craft purposeful prompts and critically evaluate the resulting output. Similarly, Lucas et al. (2024) found that teachers who received formal training in AI-assisted teaching were more confident in utilizing AI tools, resulting in a more effective integration of AI into their pedagogical practices. These findings underscored the value of targeted professional development, helping teachers harness the potential of AI in ELT. Conversely, educators with limited exposure to AI reported difficulties in formulating prompts, assessing AI-generated content, and aligning AI responses with curriculum requirements (Cacho, 2024). These educators also expressed uncertainty about the reliability of AI outputs and their adaptability to diverse learning needs, highlighting a gap between AI's potential and its practical implementation in the classroom.

Several studies have highlighted the benefits of AI in ELT when appropriately applied. Fitria (2023) found that AI played a significant role in creating a supportive learning environment for English language learners. AI facilitates personalized learning experiences, enabling students to practice English language skills tailored to their proficiency level, professional needs, and interests. A qualitative study by Aljohani (2021) examined teachers' and students' perceptions of the impact of AI on English language learning in Saudi Arabia. The findings revealed that both teachers and students recognized the positive influence of AI in enhancing language learning experiences. At the same time, challenges remain, as Celik et al. (2022) conducted a comprehensive analysis exploring the

potential and the challenges of AI in teaching practices, focusing on teachers' perspectives and their roles in AI-driven education research. Their findings underscored AI's ability to enhance lesson planning, instructional delivery, and assessment while also highlighting challenges such as technical limitations, reliability concerns, and the crucial need for teacher involvement in AI development.

Overall, the literature underscored that teachers' competence in prompt engineering is an emerging and essential factor in maximizing AI's pedagogical potential. Yet, research directly examining teachers' prompt-engineering practices, like how teachers craft prompts, evaluate AI outputs, and integrate these strategies into ELT, remains scarce. Addressing this gap highlights the significance of the present study, which explored learners' prompt engineering behaviors while acknowledging the teacher's evolving role in AI-mediated language education.

Research Questions

This study aimed to explore how English language teachers perceive and utilize AI-generated content in their classrooms, with a specific focus on the strategic engineering of prompts. The study sought to answer these three research questions:

- RQ1: How does strategic prompt engineering influence the quality and relevance of AI-generated content in ELT?
- RQ2: What are the perceptions and challenges of English language teachers in utilizing AI-generated content, and how can prompt engineering address these challenges?
- RQ3: To what extent can AI-generated content be personalized through effective prompt engineering to meet diverse student learning needs in English language classrooms?

METHODOLOGY

This study employed an explanatory sequential mixed-methods approach, beginning with the collection and analysis of quantitative data through questionnaires, followed by qualitative interviews, to further explore and contextualize the findings. The quantitative phase identified broader trends in AI usage and the effectiveness of prompt engineering among English language teachers. In contrast, the qualitative phase offered in-depth insights into teachers' experiences, strategies, and perceptions. The use of a mixed-methods approach was well-suited to this study, as it facilitated a nuanced understanding of the complex interplay between teacher practices and AI-assisted instruction. By integrating statistical trends with rich, descriptive data, this design ensured a comprehensive and well-rounded exploration of how prompt engineering could enhance AI-assisted language teaching.

Participants

This research study involved 77 English language teachers who completed a structured questionnaire and 10 English language teachers who participated in semi-structured interviews. For interviews, participants were selected through purposive sampling to ensure representation from a set of diverse teaching backgrounds, including various educational institutions, teaching levels, and degrees of experience with AI. This study was conducted with ethical approval from the Arab University. Informed consent was obtained from all participants prior to their involvement in the interviews. All data collected was treated with strict confidentiality and was not accessible to anyone. The demographic details collected included age, gender, teaching experience, educational institutions, educational qualifications, familiarity with AI tools, and proficiency in prompt engineering. This information is presented in Table 1, along with its significance value of < 0.05.

Table 1. Demographics of Participants

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	25	32.5
	Female	52	67.5
Age	25-35	15	19.5
	36-45	34	44.2
	46-55	23	29.9
	55+	05	6.5
Institution Type	Public university	45	58.5
	Private university	32	41.5
Rank	Professor	41	53.2
	Associate professor	20	25.9
	Assistant professor	16	20.7
Teaching Experience	0–5	14	18.4
	6–10	14	18.4
	11–15	19	25
	16+	29	38.2
Academic Discipline	English language	18	23.7
	English literature	20	25.3
	Linguistics	09	11.8
	Applied linguistics	09	11.8
	TEFL	05	6.6
	ELT	06	7.1
	Other	07	9.2
Familiarity With AI Tools	Highly familiar	44	57.1
	Moderately familiar	11	14.2
	Slightly familiar	13	16.8
	Not familiar	09	11.6

Note: AI = artificial intelligence; ELT = English language teaching; TEFL = teaching English as a foreign language.

The qualitative methodology is founded on the concept that the investigator aims to learn, understand, gain insight, and learn as much as possible (Merriam, 2001: 61). Moreover, purposive sampling also reflects the average person, situation, or instance of the phenomenon of interest. In this situation, purposive sampling was deemed appropriate, as it was necessary for the chosen participants to provide all the data required to answer the research study's questions. This was supported by who claimed that the primary goal of sampling is to identify individuals who can provide rich and diverse insights into the phenomenon under investigation, thereby maximizing the amount of information gathered.

Instruments

A structured questionnaire was designed by the researcher, drawing inspiration from existing literature on AI in education and ELT, but tailored specifically to the study's objectives, including assessing teachers' perceptions of AI-generated content, challenges of utilizing AI, and the effectiveness of prompt engineering. The questionnaire was self-constructed and informed by key themes identified in previous studies (Celik et al., 2022; Kumar et al., 2024). To ensure clarity and content validity, the instrument was reviewed by two experts in educational technology and language teaching and then pilot-tested with five English language instructors, resulting in minor revisions to wording and structure. The questionnaire consisted of different sections, including demographic information; awareness of and usage of AI prompt engineering; the impact of prompt engineering in ELT; challenges and ethical considerations; and future perspectives. A five-point Likert scale was used for most items, along with a few open-ended questions that allowed respondents to elaborate on their experiences and provide suggestions. Data collection was conducted over a two-month period, with the questionnaire distributed electronically via Google Forms to ensure accessibility and ease of participation. For the qualitative part, to gain deeper insights, semi-structured interviews were conducted with 10 English language teachers. Each interview lasted approximately 10 to 15 minutes and was conducted online to accommodate participants' availability and ensure contextual depth. The researcher developed the interview protocol, as seen in Table 2, based on the themes from the questionnaire and relevant literature. Questions were open-ended and allowed flexibility in wording and sequence to encourage deeper exploration of participants' views. The researcher preferred the semi-structured interview because it allowed for complete freedom to explore the reasons for digression from the original question (Cohen, 2013). As a result, participant interviews were chosen as a valuable strategy to employ. The primary reason for employing interviews was to enhance the quality and authenticity of the data.

Data Analysis

Quantitative data from the questionnaire was analyzed using IBM's Statistical Package for Social Sciences software (version 22.0). Descriptive statistics (e.g., means, standard deviations, frequencies) were used to summarize participant responses. In contrast, inferential statistics (e.g., Chi-square tests or Pearson correlations) were employed, as needed, to explore relationships between variables. Graph Pad Prism was used to create a visual representation of the data, including bar charts and graphs, to enhance interpretability. For qualitative data, semi-structured interviews were facilitated, allowing the researcher to adjust the order of questions to gain deeper insights into the participants' perspectives. Consequently, participant interviews were selected as an effective technique to implement. For this study, to account for the potential that participants attended the prearranged meeting days, personalized consent letters were sent via WhatsApp to the participants for signature prior to the interview, ensuring their voluntary participation. Additionally, telephone calls were made one day before the interview sessions to encourage attendance on the scheduled interview days. The interview lasted between 10 to 15 minutes. The optimal approach for data collection was to record the interview, contingent upon the respondent's consent (Best and Kahn, 1998). By this recommendation and the necessity to document every aspect of the interviews, the researcher recorded these sessions using a mobile application called Otter. However, the participants' approval and consent were required before the interview sessions took place. The consent letter is represented in Appendix B.

The qualitative data from the interviews were transcribed and analyzed using a thematic analysis approach. Key themes related to teachers' perceptions, challenges, and strategies were identified. The researcher used NVivo 15 software to categorize responses into main themes (parent nodes) and sub-themes (child nodes).

To enhance the trustworthiness and reliability of the qualitative data, the coding framework and identified themes were independently reviewed by two language experts at University X to enhance the validity and reliability of the questions. Their feedback was used to refine theme

definitions and address any discrepancies. Inter-rater reliability was ensured through structured discussions, during which consensus was reached on all coding decisions, thereby confirming the validity and consistency of the thematic structure. As indicated above, predefined themes recognized in the literature were employed based on theory or themes that developed and emerged from the raw data (Miles & Huberman, 1994). NVivo 15 software, on the other hand, can manage, through its innate capacity, to treat audio or video captured to satisfy the needs of researchers because it has simple methods of submitting questions and collecting knowledge of the research phenomena, and acquires coding comparative ability that enables researchers to assess the coding process reliability (Ishak & Bakar, 2012).

All related textual data was coded with each corresponding "node/theme." Parent nodes (main themes) and child nodes (sub-themes) were established to examine teachers' approaches to prompt formulation, the challenges they encountered, and the benefits they perceived in using AI-generated materials, as observed in the audiotaped recordings. Subsequently, all pertinent nodes/themes were organized beneath each corresponding parent node. The coding procedure used in the current research was executed utilizing the coding method proposed by Strauss and Corbin (1998), which encompassed: a) open coding, b) axial coding, and c) selective coding. The researcher adhered to the guidelines for the interview method established by Thomas (2004) and Daniels and Cannice (2004). The table in Appendix A delineates the progression of the interview procedure.

RESULTS

Data Collected Through Questionnaires

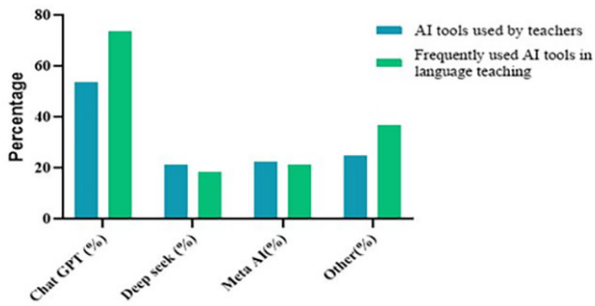
Demographic Profile of Participants

A total of 77 English language teachers completed the questionnaire, while 10 teachers participated in semi-structured interviews. The participants represented a diverse age range, categorized as follows: 25–35 years, 36–45 years, 46–55 years, and over 55 years. The gender breakdown comprised 32.5% male and 67.5% female respondents. Participants came from both public universities (58.5%) and private universities (41.5%). Their academic ranks ranged from professor (53.2%) and associate professor (25.9%) to assistant professor (20.7%). Teaching experience was another key variable, categorized into 0–5 years (18.4%), 6–10 years (18.4%), 11–15 years (25%), and more than 15 years (38.2%), offering insight into how professional experience influences AI adoption. The academic disciplines represented included English language, literature, linguistics, applied linguistics, TEFL, ELT, and other related fields. Participants were also asked to report their familiarity with AI tools (e.g., ChatGPT, Bard, and other platforms), categorized as highly familiar (57.1%), moderately familiar (14.2%), slightly familiar (16.8%), and not familiar (11.6%). This demographic analysis provided essential context for understanding the perspectives and experiences of the participants regarding AI prompt engineering in ELT. This information is presented in Table 1 along with its significance value.

Role of Prompt Engineering in Teaching With AI Tools

In the evolving landscape of education, AI tools such as ChatGPT, DeepSeek, and MetaAI are gaining popularity for helping teachers deliver personalized, efficient, and innovative lessons. Prompt engineering plays a crucial role in ensuring that these AI tools are utilized effectively, enabling teachers to tailor responses, create dynamic lesson plans, and provide personalized feedback. As shown in Figure 1, ChatGPT emerged as the most frequently used AI tool, significantly surpassing others such as DeepSeek and MetaAI. This trend indicates that teachers tend to favor well-established platforms, likely due to their accessibility, widespread visibility, and overall familiarity. In contrast, AI tools with more specialized functions or limited exposure were considerably less utilized.

Figure 1. Evaluating the Role of AI Tools in Language Teaching Through Teachers' Perceptions

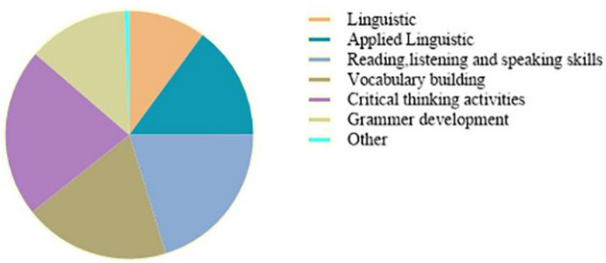


Based on survey responses and graphical data representations, the graph explored key areas, such as the usage and frequency of AI tools. The analysis reflected teachers' insights into the effectiveness, usage, and frequency of AI tools in language education. This trend suggested that teachers tend to favor well-established platforms, likely due to their accessibility, widespread visibility, and overall familiarity. In contrast, AI tools with more specialized functions or limited exposure were considerably less utilized.

Utilization of AI Prompt Engineering in Language Teaching

This section responds to RQ2 by identifying the diverse learning needs of students, specifically in vocabulary development, grammar support, critical thinking, communicative competence, and comprehension and demonstrates how teachers utilize prompt engineering to address these needs. The research examined how AI prompt engineering is employed by teachers in language teaching, with a specific focus on various aspects of language development, including vocabulary building, grammar development, critical thinking, and communicative skills. By creating tailored prompts for AI tools, teachers can enhance their instructional methods and foster more effective learning outcomes. A pie chart was created to visually represent the frequency and distribution of AI tool usage among teachers across different areas of language learning. Figure 2 presents this distribution, highlighting the primary focus areas in language learning activities.

Figure 2. Utilization of Prompt Engineering Across Language Teaching Domains



A significant portion was devoted to reading, listening, and speaking skills, followed closely by critical thinking activities and vocabulary building. This also indicated a strong emphasis on both communicative competence and cognitive engagement. Grammar development also represented a substantial portion, underscoring its sustained importance in language

instruction. In contrast, areas such as language theory, discourse analysis, and other specialized domains were minimally represented, suggesting a relatively lower prioritization of these advanced or theoretical components in the curriculum. These patterns suggest that students' learning needs are diverse, spanning foundational skills (vocabulary, grammar), cognitive skills (critical thinking), and communicative abilities, all of which teachers can address through carefully designed AI prompts that scaffold explanations, generate examples, and support practice-based learning, thereby directly answering RQ2 and RQ3.

Effectiveness and Classroom Integration of AI-Generated Content

Effectiveness and integration of AI-generated content in classroom teaching, with a focus on key aspects, includes lesson planning, student engagement, comparison between traditional and AI-generated content, and modification of AI-generated content. To address RQ3, this section examined how prompt engineering enhanced lesson planning, student engagement, and instructional effectiveness. The findings were analyzed based on the percentage of teachers who agreed with each aspect, the sample size (n), the Chi-square test statistic (X^2), degrees of freedom (df), and the corresponding p-values. This is depicted in Table 2.

Table 2. Teachers' Perceptions on the Effectiveness and Integration of AI Tools in Language Teaching

Aspects	% Agree	n	X^2 (df)	P Value
Lesson Planning	68%	60	10.56 (1.63)	0.001
Student Engagement	55%	47	11.43 (2.55)	0.003
Traditional Versus AI-Generated Content	38%	33	3.02 (1.53)	0.082
Modification of AI Content	46%	41	4.87 (1.8)	0.067

Note. AI = artificial intelligence; df = degrees of freedom; n = sample size; X^2 = Chi-square test statistic.

As shown in the table, teachers expressed the highest levels of agreement with the positive impact of AI on lesson planning (68%) and student engagement (55%). Chi-square tests indicated that both aspects were statistically significant ($p = 0.001$ and $p = 0.003$, respectively), suggesting that AI tools are viewed as effective in supporting instructional preparation and promoting learner interaction. However, perceptions regarding the use of AI-generated content in comparison to traditional materials (38%) and its modification (46%) were lower and did not reach statistical significance ($p = 0.082$ and $p = 0.067$, respectively). These results reflect ongoing teacher skepticism or uncertainty about fully integrating AI-generated content into traditional classroom practices. Overall, the data indicated that, while teachers acknowledge AI's utility in certain pedagogical areas, broader acceptance and integration remain tentative.

Challenges and Ethical Considerations in AI-Generated Content

The study investigated educator concerns regarding AI-generated content through a systematic evaluation of three key dimensions, as presented in Table 3. The analysis revealed significant patterns in teacher perceptions of challenges and ethical considerations when utilizing prompt engineering in pedagogical contexts.

Table 3. Teachers' Concerns Regarding the Use of AI Tools in Language Teaching

Concern	% Reported	n	X^2 (df)	P Value	Interpretation
Accuracy	49%	43	$\chi^2 = 1.02$	0.312	Not significant
Plagiarism	38%	33	z = 2.18	0.029	Statistically significant
Institutional Restriction	45%	39	$\chi^2 = 3.67$	0.055	Marginal significant

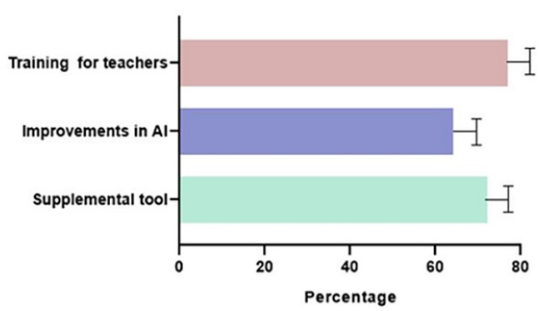
Note. AI = artificial intelligence; df = degrees of freedom; n = sample size; X^2 = Chi-square test statistic.

Table 3 presents the primary concerns reported by language teachers regarding the use of AI tools, including accuracy (49%), plagiarism (38%), and institutional restrictions (45%). A Chi-square test revealed that concerns about accuracy were not statistically significant (χ^2 (1) = 1.02, p = 0.312), suggesting that such concerns were not concentrated in any specific subgroup of respondents. In contrast, plagiarism concerns were statistically significant (z =2.18, p = 0.029), indicating that apprehension about the potential misuse of AI-generated content was prevalent among the respondents. Concerns about institutional restrictions approached significance (χ^2 (1) = 3.67, p = 0.055), suggesting a possible trend worth further investigation. These findings highlighted varying levels of concern, with plagiarism emerging as a statistically significant issue. At the same time, accuracy and institutional barriers, though present, did not reach the same level of statistical significance.

Future Prospects and Recommendations for AI in Language Teaching

As AI technologies become increasingly embedded in educational ecosystems, this study identified three critical pathways to optimize their implementation in language instruction: targeted teacher training, the refinement of AI tools, and strategic integration as a supplemental tool in educational settings. Figure 3 serves as a foundation for these insights, illustrating educators' expectations and priorities for the future use of AI in language education.

Figure 3. Key Recommendations for Effective Integration of AI in Language Teaching



The graph reveals a strong consensus among respondents on the future directions for integrating AI in education. Training for teachers emerged as the top priority, suggesting a recognized need for capacity building to use AI tools effectively. Meanwhile, there was also considerable support for leveraging AI as a supplemental tool and for ongoing improvements in AI, indicating that, while educators value the technology, they also see room for enhancement and structured implementation.

Thematic Analysis of Qualitative Data on Prompt Engineering in ELT.

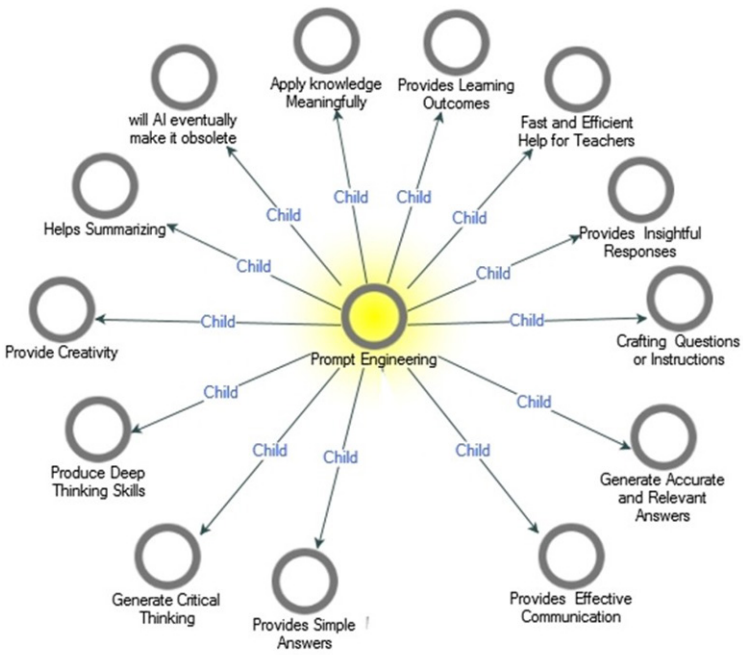
Prompt engineering was identified as a central theme that contributed to teachers teaching at different university levels in the English department. The analysis of teacher interviews revealed nine key themes related to the use of prompt engineering in AI-assisted English language instruction.

Analysis of teacher interviews revealed sub-theme categories in the context of AI-assisted language teaching. Teachers emphasized that effective prompts help generate accurate, relevant, and time-efficient content for lesson planning. AI tools support simplification and summarization of material, offering direct benefits for both teaching and learning. As Teacher 5 noted, "AI can summarize or give simple answers for students and teachers," while Teacher 4 described it as "fast and efficient in helping the teachers." Many teachers highlighted the ability of prompt engineering to foster deeper learning, creativity, and critical thinking. For instance, Teacher 7 remarked, "an effective and accurate Prompts that really get students thinking deeply and creatively," while Teacher 9 connected the simple and relevant prompts to "critical thinking, creativity, and effective communication."

Participants also expressed concerns about AI's accuracy, potential over-reliance, and alignment with curriculum standards. They emphasized the need for teacher training in prompt crafting to ensure pedagogical alignment. Teacher 3 summarized its dual role by saying, "Prompt Engineering can help teachers in crafting questions or instructions that guide students to generate accurate, relevant, and insightful responses."

Figure 4 below is the detailed presentation of themes supported by interview data and corresponding figures.

Figure 4. Prompt Engineering



AI as a Complementary Teaching Tool

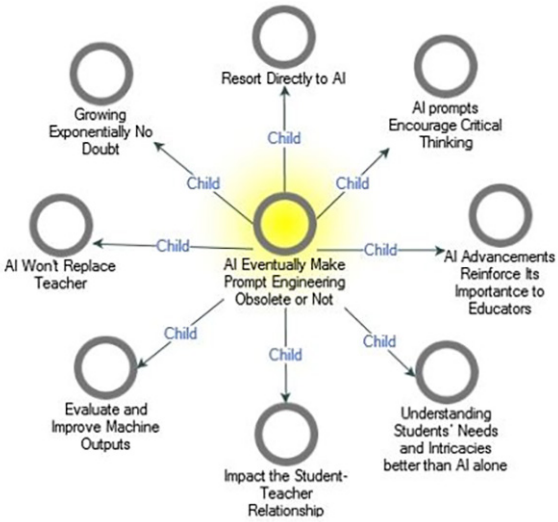
The interviews revealed that teachers broadly viewed AI as a supportive and complementary tool in academic settings rather than a replacement for educators. Participants acknowledged AI's supportive role, particularly in enhancing teaching efficiency, personalizing content, and assisting with classroom activities. For instance, Teacher 1 stated, "AI developments highlight its rising

importance for teachers by improving teaching competence, personalization, and student engagement." However, participants also emphasized that the effectiveness of AI depends on the quality of prompts provided by teachers. In this sense, prompt engineering emerged as a crucial practice that determines how AI tools generate relevant, accurate, and pedagogically useful content. Teacher 8 highlighted this distinction by noting, "AI, directed by considerate instructor prompts, improves knowledge but cannot fully replace a teacher's deep understanding of students' needs." This reflects the understanding that while AI offers powerful capabilities, it is prompt engineering that enables teachers to guide AI toward meaningful learning outcomes. Teachers widely insisted on the irreplaceable role of human educators in addressing students' emotional and intellectual needs. As Teacher 6 explained, "It helps as an influential teaching aid rather than replacing teachers." The findings suggested that AI is best understood as a complementary tool, with prompt engineering serving as the bridge that aligns AI outputs with classroom objectives and preserves the central role of the teacher.

Figures 5 through 13 illustrate the list of relevant sub-themes extracted from NVivo software.

Figure 5 illustrates the main theme and sub-themes of AI as a complementary tool in academic settings.

Figure 5. AI as a Complementary Teaching Tool



Limitations and Challenges

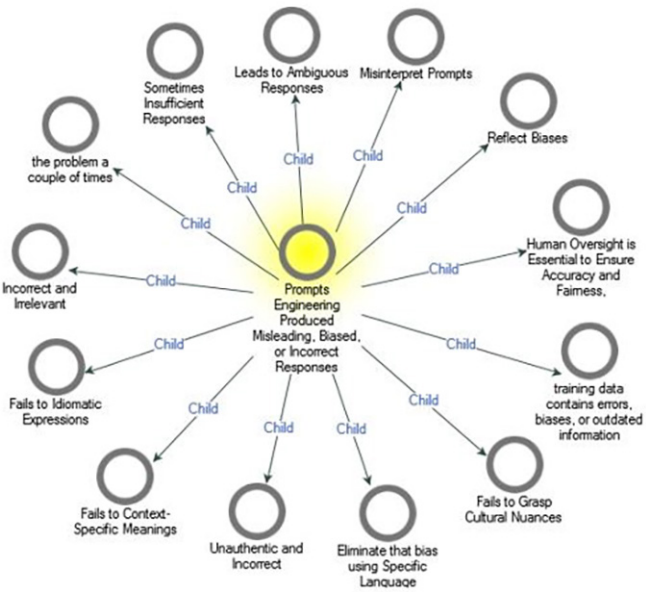
Analysis of participant responses revealed several concerns related to the implementation and reliability of AI tools in educational settings. The sub-themes under this category suggested that prompt engineering can sometimes lead to misleading, biased, or incorrect responses. Several participants noted that AI often fails to grasp context-specific meanings and idiomatic expressions, which can affect the clarity and relevance of its output. As Teacher 2 remarked, "The process of AI is a language based on patterns, but sometimes it has deficiencies in deep contextual understanding. Also, AI makes it difficult to interpret idioms and nuanced senses correctly."

Concerns were raised about the reliability and factual accuracy of AI-generated content. Teacher 5 reported experiences: "Sometimes it gives incorrect, irrelevant responses," and Teacher 7 also highlighted issues: "Often, it provides unauthentic and incorrect responses" (in the generated outputs), indicating the potential for misinformation if not carefully monitored. Teachers 4 and 8 emphasized the importance of maintaining human

involvement when using AI in academic settings. Teacher 9 stated, "Human oversight is essential to ensure accuracy and fairness," underscoring the belief that AI should function as a support tool rather than a standalone solution.

Figure 6 represents the theme and sub-themes of prompt engineering that produced misleading, biased, and incorrect responses.

Figure 6. Biased, Misleading, and Incorrect Responses

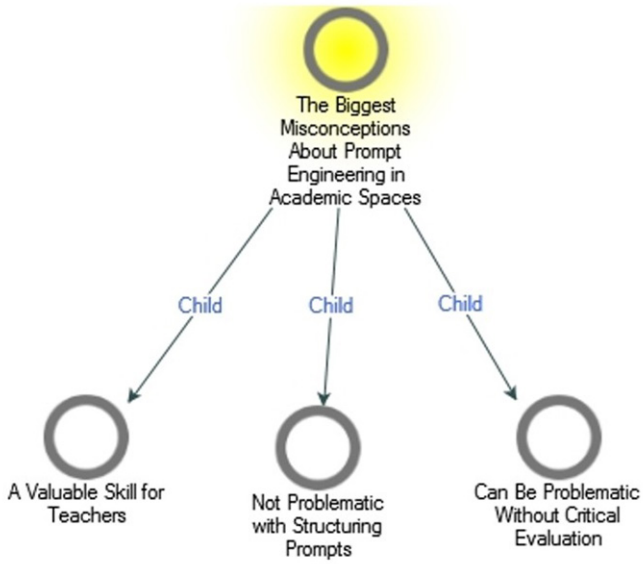


Clarifying Misconceptions and Exploring Academic Applications

Participants emphasized the importance of addressing common misconceptions about AI use in education, while also acknowledging its increasing utility in academic research contexts. The sub-themes highlighted the misconceptions surrounding prompt engineering in academia. Several participants emphasized that effective AI usage (particularly in prompt engineering) is a skill that educators must develop. Teacher 3 noted, "It's a valuable skill for teachers in the language learning process," while Teacher 1 clarified, "I realized that the problem wasn't with the AI itself, but with how I was structuring my prompts." These reflections suggested that misconceptions often stem from user inexperience rather than the limitations of the AI itself.

Figure 7 illustrates the main theme and sub-themes of the most significant misconception in prompt engineering in academic settings.

Figure 7. The Biggest Misconception about Prompt Engineering

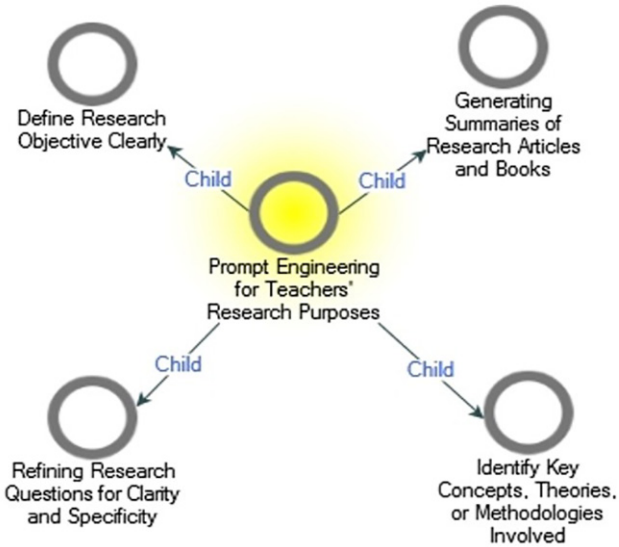


Beyond classroom support, participants also identified the AI's potential to contribute to scholarly or research work. Teacher 4 shared that AI defines prevalent beliefs regarding prompt engineering in academic contexts, portraying it as either a beneficial pedagogical skill, unproblematic when prompts are meticulously structured, or possibly problematic in the absence of critical analysis.

It emphasizes that prompt Engineering is not intrinsically detrimental; rather, its effects are contingent upon the degree of thoughtfulness and criticality in its application within educational settings." Teacher 7 added, "AI can create applicable philosophies, ideas, and methods based on well-designed prompts, rationalizing the research procedure." These insights point to AI's evolving role in supporting both pedagogical and research-related tasks.

Figure 8 illustrates the main theme and sub-themes of prompt engineering used by language teachers for research purposes.

Figure 8. Prompt Engineering for Research Purposes

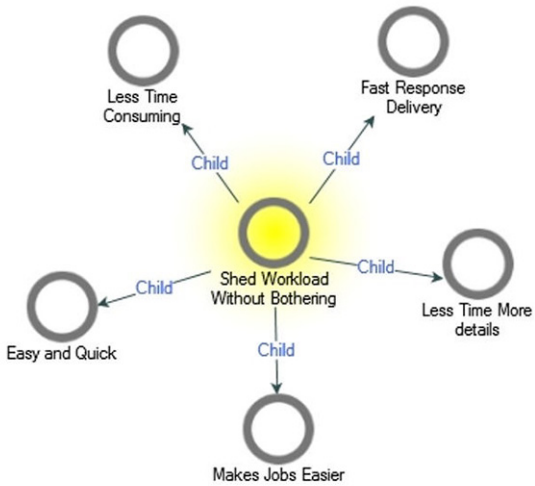


Enhancing Efficiency and Supporting Language Learning

Participants identified several practical benefits associated with AI integration in educational contexts; however, they emphasized that these benefits emerged specifically when teachers applied deliberate prompt-engineering strategies. Rather than "streamlining tasks" on its own, prompt engineering enabled teachers to guide the AI toward producing outputs that were more accurate, relevant, and instructionally useful. Teachers explained that by designing precise prompts, such as "Summarize this passage for intermediate English as a second language learners using simple sentences," "Generate a lesson outline on the past tense focusing on form, use, and examples," or "Provide five simplified explanations of this concept suitable for low-proficiency learners," they could obtain well-structured materials that significantly reduced preparation time. In this way, it was the teacher's prompt-design choices that streamlined academic tasks, not prompt engineering as an abstract concept.

Figure 9 illustrates the main theme and its sub-themes with a focus on reducing the daily workload of language teachers.

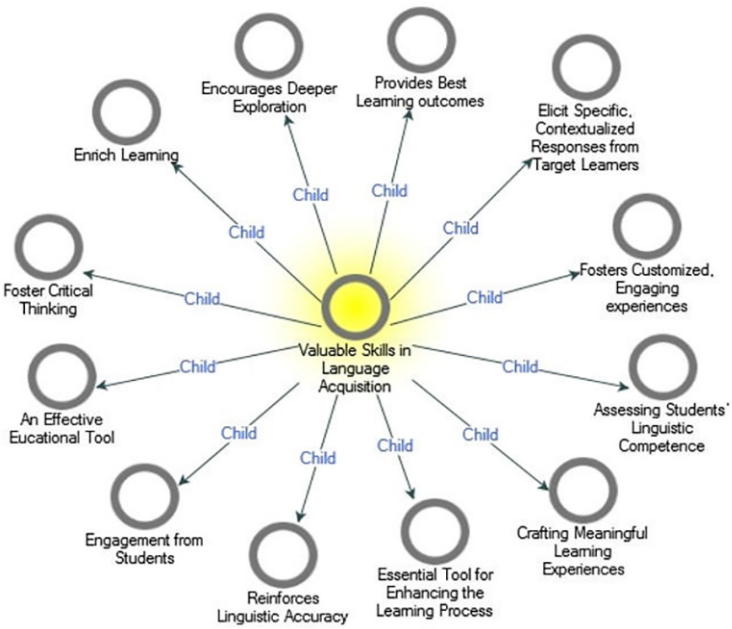
Figure 9. Shed Workload Without Bothering



This theme addressed RQ2 by demonstrating how prompt engineering enables teachers to obtain simplified explanations, vocabulary lists, structured lesson plans, grammar corrections, and immediate feedback tailored to learners' needs. As Teacher 7 and Teacher 2 noted, using well-designed prompts "speeds up the writing of our work and we can write a literature review in less time," because the system quickly recommends related theories, methods, and literature. Teacher 5 similarly explained that AI becomes more effective "when we give precise instructions," noting that clear prompts allow the tool to refine the content, summarize long passages, and reduce the manual effort involved in lesson planning. Teachers also noted that once appropriate prompts were crafted, the resulting AI-generated materials supported language learning more effectively. For instance, Teacher 8 highlighted that "[prompt engineering] helps in vocabulary, and sentence structure in writing and grammar check," while Teacher 6 observed that "[prompt engineering] delivers instant student responses and keeps students involved in class discussion." These examples demonstrated that the pedagogical value stems from how teachers craft their prompts, rather than from the AI tool operating independently.

Sub-themes related to prompt engineering as a valuable skill in language learning are depicted in Figure 10, which illustrates the main theme and sub-themes of prompt engineering as a valuable skill in language acquisition for language teachers.

Figure 10. Valuable Skills in Language Acquisition

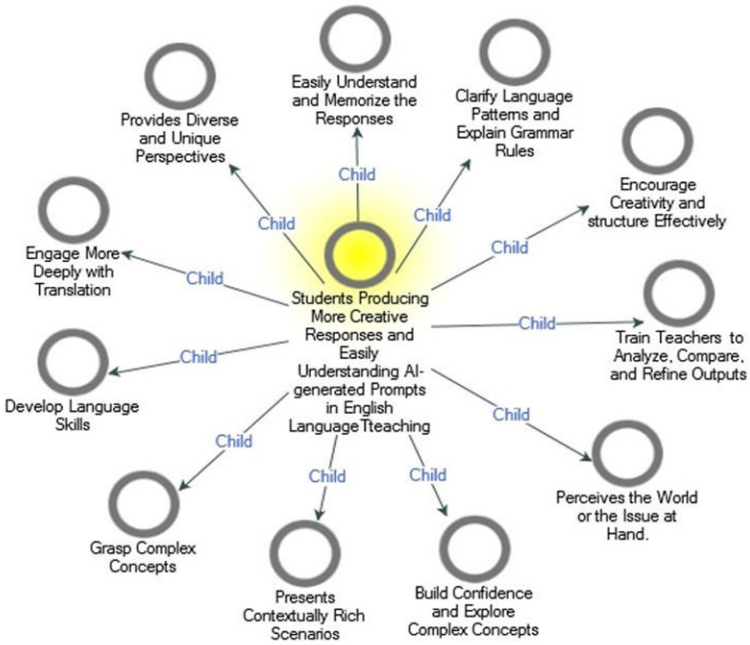


Enhancing Student Learning and Redefining Educator Roles

Addressing RQ3, the theme represented in Figure 10, illustrated how well-designed prompts facilitate students' comprehension difficulties, enhance fluency, foster confidence, and encourage deeper engagement with course content. Participants reported that prompt engineering integration had a meaningful impact on student learning outcomes while also reshaping the dynamic between educators and technology. Sub-themes under the prompt engineering–teaching content relationship revealed a notable impact on students' learning. Many participants noted that AI tools facilitated better comprehension by simplifying complex concepts. Teacher 3 noted that "[AI] shortens hard thoughts by breaking them down into clear, organized details." Furthermore, AI was seen as a confidence booster, particularly when it provided "[AI] makes accurate, context-based specimens to improve learning" (Teacher 7), and "[AI] increases students' vocabulary and fluency" (Teacher 5), which supported student learning.

Figure 11 illustrates the main theme and sub-themes of prompt engineering's role in students' performance as they produced more creative responses and easily understood the prompts in ELT.

Figure 11. Improvements in Student Performance

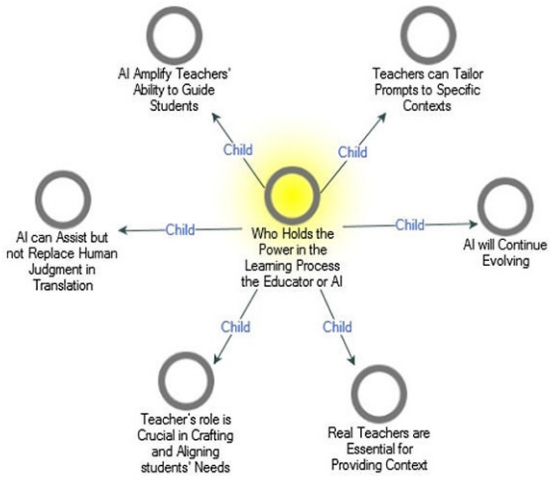


While participants recognized AI's role in enhancing instruction, they also emphasized the irreplaceable role of teachers. Teacher 9 stated, "Humans are humans. Only we as teachers can give or provide them the contextual clues, not AI," (Teacher 9), highlighting the importance of human judgment and classroom awareness. At the same time, Teacher 4 and Teacher 1 stated, "[Human judgement] amplifies their ability to guide students toward deeper, more thoughtful learning" and "In my viewpoint, we cannot totally replace [human judgement]. In fact, AI can assist but not replace human judgment in translation," suggesting a complementary relationship between educators and AI.

Sub-themes under prompt engineering revealed its emerging role as a power-holding element in the learning process, often in comparison to the traditional authority of teachers.

Figure 12 illustrates the central theme and its subthemes, as perceived by teachers, regarding prompt engineering as a factor influencing power in the learning process.

Figure 12. Power-Holding AI or Educators



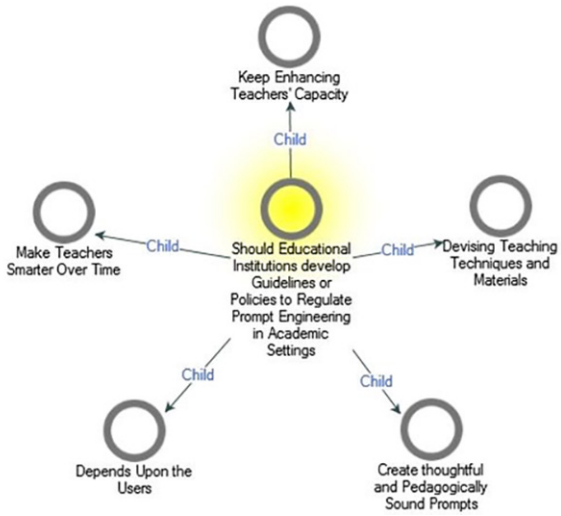
Policy Recommendations and Institutional Support

Participants emphasized the importance of institutional involvement in guiding the effective integration of AI within educational settings. Teachers 3, 7, and 2 expressed the need for structured frameworks to support prompt engineering practices. Teacher 2 highlighted the importance of creating, stating, "[Structured frameworks have] the ability to create thoughtful and pedagogically sound prompts which remain essential," suggesting that clear instructional guidelines would enhance the educational value of AI-generated content.

Participants also acknowledged the role of professional development in adapting to AI-enhanced teaching. As Teacher 5 and Teacher 6 participants noted, "AI will keep enhancing our capacity to come up with more creative and critical prompts" (Teacher 5) and "Interaction between AI and teachers will make teachers smarter over time" (Teacher 6), indicating that with proper support, educators can evolve their teaching strategies to leverage AI tools better.

Figure 13 illustrates the main theme and its sub-themes regarding the development of guidelines and policies for using prompt engineering in academic settings.

Figure 13. Guidelines and Policies Regarding AI Usage in Academics



DISCUSSION

The present study investigated how AI-generated content, when strategically engineered through effective prompt design, can enhance ELT. By integrating questionnaire data and interviews from English language teachers, this research aimed to understand how tailored prompts could enhance the quality and effectiveness of AI-generated content in an educational context. Questionnaire responses revealed that teachers appreciated AI for generating diverse exercises, simplifying lesson preparation, and offering customized materials suitable for different proficiency levels. However, interview data added depth by showing that these benefits were not automatic; rather, they emerged when teachers deliberately designed prompts with clear instructional intent. Teachers repeatedly emphasized that poorly constructed prompts, particularly those lacking specificity, often produced irrelevant, overly generic, or culturally inappropriate outputs. This resonated with prior research, yet extended the literature by demonstrating how specific prompt-engineering practices can mitigate these issues in real classroom settings (Tan et al., 2025). In this study, teachers employed three main strategies that directly influenced the quality of AI-generated materials: task-specific prompting, contextual-constraint prompting, and interactive refinement. These strategies emerged consistently across participants, demonstrating a strong empirical link between prompt design and instructional outcomes.

Teachers consistently explained that poorly constructed prompts, particularly those lacking specificity, produced outputs that were misaligned with learners' proficiency levels, too generic for lesson objectives, or culturally or linguistically inappropriate. This finding aligns with prior research (Choi & Chang, 2025; Liu et al., 2022; Yang et al., 2022); however, the present study extends the literature by demonstrating which prompt-engineering practices effectively mitigate these issues in real-world classrooms. Unlike prior work that broadly framed prompt engineering as writing "better prompts," this study demonstrated that teachers apply three specific prompt-engineering strategies that directly shape instructional outcomes like task-specific prompting, contextual constraint prompting, and interactive refinement prompting. This aligned with the research, which claimed that prompt engineering has a positive role in the teaching and learning process. This led to the conclusion that prompt engineering is not just a technical skill but also requires pedagogical knowledge to be effective in the ELT context (Hegazy, 2024; Lee et al., 2024; Walter, 2024). One of the key insights from the study was that prompt engineering should not be viewed as a mere technical activity but as an art that blends both linguistic and educational expertise. Teachers who understood the nuances of the

language, as well as the learning needs of their students, were better equipped to craft prompts that elicited high-quality AI-generated content. These teachers reported a significant improvement in classroom dynamics when AI tools were used to create customized exercises, real-world scenarios, and targeted reading materials (Vashishth et al., 2024; Yadav, 2025).

Based on the findings, teachers recommended incorporating prompt engineering training as part of professional development programs. They emphasized that understanding how to craft, test, and refine prompts is essential for maximizing the potential of AI tools in the classroom. Teachers also suggested that prompt engineering should be introduced as a supplemental teaching skill, empowering educators to generate more relevant and personalized content aligned with specific learning objectives (Cain, 2024; Mzwri & Turcsányi-Szabo, 2025). By equipping teachers with this knowledge, institutions can enhance the effectiveness of AI-generated content and ensure its alignment with diverse student needs (Almuhanna, 2024). From a personalization perspective (RQ3), this study revealed that strategic prompt engineering significantly enhances the adaptability of AI content. Teachers used prompts to tailor content for students with varying learning styles, proficiency levels, and specific objectives. Teachers may start with a general prompt, such as "Generate a short story using new vocabulary words," and then refine it to "Generate a short story set in a marketplace that uses at least five of the following vocabulary words: negotiate, bargain, vendor, purchase, and quality." Through such refinements, prompt engineering not only enhances the precision of AI-generated content but also directly addresses common classroom challenges, such as promoting contextualized vocabulary use, scaffolding critical thinking, and supporting differentiated instruction.

Despite the promising findings, the study had several limitations. One major limitation was the small sample size, as only a limited number of teachers were surveyed and interviewed. As such, the findings may not be fully representative of the broader population of English language teachers, particularly in different educational settings or regions. Another limitation was the variability in teachers' familiarity with AI tools and their ability to craft effective prompts. Some teachers had limited experience with AI, which made it challenging for them to fully leverage the potential of AI-generated content. As a result, the effectiveness of prompt engineering was influenced by the individual language teacher's technical competence and pedagogical expertise (Knoth et al., 2024; Korzynski et al., 2023).

Moreover, the study did not explore the long-term impacts of AI-generated content on student learning outcomes. While the immediate feedback from teachers was positive, the sustainability and lasting effects of AI tools on language acquisition and teaching effectiveness need further exploration (Al-Smadi et al., 2024; Wei, 2023). The research opens several avenues for future work in the field of AI-enhanced language teaching. One promising direction is the development of training programs for teachers to enhance their understanding and effective implementation of strategic prompt engineering. Such programs could focus on enhancing language teachers' technical skills in using AI tools while also emphasizing the pedagogical principles of prompt crafting (Ajani, 2024; Wu, 2024). Another potential area for exploration is the integration of AI with other educational technologies, such as adaptive learning systems, to create a more comprehensive and personalized learning environment. Overall, the findings suggested that effective prompt engineering plays a crucial role in determining the instructional value of AI in ELT and should be regarded as an essential skill for contemporary language teachers.

CONCLUSION

This study underscored the transformative potential of AI-generated content in English language classrooms, particularly when driven by strategic and well-informed prompt engineering. The study demonstrated that practical prompt engineering significantly enhances the utility of AI-generated content in ELT. Specifically, it revealed that carefully constructed and iteratively refined prompts lead to more relevant, personalized, and engaging learning materials tailored to the diverse needs of learners.

Teachers indicated that AI tools significantly enhanced the efficiency of lesson planning and fostered greater student engagement, serving as supportive complements rather than substitutes for human instruction. However, several challenges hinder the optimal integration of AI. Chief among them is the lack of formal training in prompt design and limited technical confidence among teachers. These barriers must be addressed through targeted professional development programs that equip educators with practical skills in AI prompt engineering and critical evaluation of AI outputs. Implications for practice include incorporating hands-on AI workshops into teacher education curricula, encouraging collaborative development of prompts among teaching staff, and providing institutional support through policy frameworks that guide the ethical and practical use of AI in classrooms. Future studies should investigate the long-term effects of AI-generated content, compare AI-assisted instruction with traditional instruction, and assess the impact of prompt engineering training on teaching efficacy and student achievement. These directions will deepen our understanding of AI's evolving role in pedagogy and ensure its responsible and impactful adoption in English language education.

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COMPETING INTERESTS STATEMENT

The author of this publication declares there are no competing interests.

CORRESPONDING AUTHOR

Correspondence should be addressed to Aisha Bhatti: a.bhatti@psau.edu.sa

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APPENDIX A

The Flow of the Interview Process

Sequence	Steps in the Interview Process
1	**Opening Phase** ☐ Greeting ☐ Introduce yourself ☐ Establish port by discussing general matters ☐ Describe the study, the purpose, and the importance of the research ☐ Appreciation for the willingness to participate ☐ Confirmation of confidentiality and anonymity ☐ Grant permission for tape recording of the interview
2	**Question-Answer Phase** ☐ Questioning and probing: followed the interview protocol ☐ Listening: practice 'active listening' throughout the session ☐ Recording: took notes on pertinent things mentioned or noticed during the session ☐ Maintaining rapport: sustain a positive climate throughout the sessions and keep the interview 'in play.'
3	**Closing Phase** ☐ Clarifying: asked whether any points needed to be clarified ☐ Thanking: thanks for cooperation ☐ Follow-up: for further cooperation

Note: Source adapted from Thomas (2004) and Daniels and Cannice (2004)

APPENDIX B

Interview Consent Letter

Strategic Prompt Engineering for Enhancing AI-Generated Content in ELT: Empowering EFL Contexts

The following information is provided for you to decide whether you wish to participate in the present study. The purpose of this study is to investigate how English language teachers at a Saudi university utilize AI tools for lesson planning and classroom activities. Through interviews and questionnaires, the study will examine teachers' approaches to formulating prompts, the challenges they encounter, and the benefits they perceive from using AI-generated materials. Do not hesitate to ask any questions about the study either before participating or during the time you are participating. I would be happy to share the findings with you after the research is completed.

My consent: I agree to take part in the above research project. I have been informed about the project and have read the Explanatory Statement, which I will keep on file for my records. I understand that agreeing to take part means that:

Please tick the appropriate box:

- I am willing to be interviewed by the researcher.
- I allow the interview to be audio taped.
- I make myself available for a further interview should that be required.
- I also understand that my participation is voluntary, that I can choose not to participate in part or the entire project, and that I can withdraw at any stage of the project without penalty or disadvantage.

Please note that this consent form is a completely confidential document. It will be used solely for the researcher's project.

Please sign your consent with full knowledge of the nature and purpose of the procedure.

Signature of Participant: _____
Designation: _____
University/College: _____
Email: _____
Date: _____

Dr. Aisha Bhatti serves as an assistant professor in the Department of English, Prince Sattam bin Abdulaziz University in Saudi Arabia. Dr. Bhatti served as a reviewer for conference papers for the Linguistics Society of America. She is presently serving as an editorial board member for the World Journal of the English Language and as a reviewer for the MEXTESOL Journal. Her research publications demonstrate her proficiency in bilingualism, language change, code-switching, and second language acquisition. She is actively involved in professional societies, including the Saudi Reading Society, KSAALT, and the Society of Pakistani English Language Teachers, all of which are dedicated to advancing education and research in her discipline.

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